Advanced Derivatives Assignment 1

Léon Le Bohellec | 342583,
Arthur Stämpfli | 396577,
Kilian Wan | 340838

September 29, 2025

Exercise 1

We know from the lectures that the price of the call in the jump-diffusion case at $t = 0$ is given by the following :

$$C(S_0) = \sum_{j \geqslant 0} \mathbb{P}(N(T) = j) C^{BS}\left(S_0(1-\gamma)^j, K, T, \sigma, r, q - \lambda\gamma\right).$$

However, this is whenever the (risk neutral) dynamics of the stock are the following :

$$\frac{dS_t}{S_t} = (r - q + \lambda\gamma)dt + \sigma dW^{\mathrm{Q}}(t) - \gamma dN(t).$$

We observe the minus sign in front of the jump process part. But in our case, the dynamics under the risk neutral measure are given by

$$\frac{dS_t}{S_t} = (r - \lambda^Q\gamma)dt + \sigma dW(t) + \gamma dN(t),$$

with $r = 0.04$, $\sigma = 0.2$, $\lambda^Q = 0.2$ and $\gamma = -0.8$. We decide to recompute the call option price on a stock with such dynamics. Using Ito's formula for jump processes, we can get the dynamics of $\log(S_t)$ as:

$$d\log(S_t) = \frac{\partial \log(S_t)}{\partial S_t} dS^c + \frac{1}{2}\frac{\partial^2 \log(S_t)}{\partial S_t^2}(dS_t^c)^2 + \left(\log(S_t^+) - \log(S_t)\right) dN(t)$$

where S_t^c is the continuous part of S_t, $S_t^+ = (1+\gamma)S_t$, the jump in the stock price due to a jump at time t, $\partial \log(S_t)/\partial S_t = 1/S_t$, $\partial^2 \log(S_t)/\partial S_t^2 = -1/S_t^2$ and $(dS_t)^2 = S_t^2\sigma^2 dt$. We thus have:

$$d\log(S_t) = \left(r - \lambda^Q\gamma - \frac{1}{2}\sigma^2\right) dt + \sigma dW(t) + \log(1+\gamma)dN_t$$

We can have an explicit formula for S_T:

$$S_T = S_0(1+\gamma)^{N_T} e^{(r-\lambda^Q\gamma-\frac{1}{2}\sigma^2)T+\sigma W(T)}$$

We are under the risk neutral measure, and thus the price of a call option is the expectation of the discounted payoff:

$$\begin{aligned}
C^j(S_0) &= \mathrm{E}\left[e^{-rT}(S_T-K)^+\right] \\
&= \mathrm{E}\left[e^{-rT}\left(S_0(1+\gamma)^{N_T}e^{(r-\lambda^Q\gamma-\frac{1}{2}\sigma^2)T+\sigma W(T)}-K\right)^+\right] \\
&= \sum_{j=0}^{\infty} P(N_T=j)\mathrm{E}\left[e^{-rT}\left(S_0(1+\gamma)^{N_T}e^{(r-\lambda^Q\gamma-\frac{1}{2}\sigma^2)T+\sigma W(T)}-K\right)^+ \middle| N_T=j\right] \\
&= \sum_{j=0}^{\infty} e^{\lambda^Q T}\frac{(\lambda^Q T)^j}{j!}\mathrm{E}\left[e^{-rT}\left(S_0(1+\gamma)^{j}e^{(r-\lambda^Q\gamma-\frac{1}{2}\sigma^2)T+\sigma W(T)}-K\right)^+\right] \\
&= \sum_{j=0}^{\infty} e^{\lambda^Q T}\frac{(\lambda^Q T)^j}{j!}C^{BS}(S_t=S_0(1+\gamma)^j, T, K, \sigma, q=\lambda^Q\gamma), \qquad \text{(BS)}
\end{aligned}$$

where C^{BS} is the Black-Scholes and Merton call option with initial price $S = S_0(1+\gamma)^j$, for a stock paying dividends at constant rate $q = \lambda^Q\gamma$. Recall the BS formula:

$$C^{BS}(S_0, T, K, \sigma, q) = S_0 e^{-qT}\Phi(d_1) - Ke^{-rT}\Phi(d_2)$$

with

$$d_1 = \frac{\log\left(\frac{S_0}{K}\right) + \left(r - q + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} \quad , \quad d_2 = d_1 - \sigma\sqrt{T}$$

In order to implement this infinite sum, we used a function that finds the value n such that $P(N(T) < n) \geqslant 1 - 10^{-14}$. This function is written as a pseudo in **Algorithm** 1, and allows us to stop the sum as soon as most of the mass of probability has contributed.

Algorithm 1 Find minimal n such that coverage $\geqslant 1 - 10^{-14}$

Input: t, λ
Output: n
$x \leftarrow 0 \quad n \leftarrow 0$
while $x < 1 - 10^{-14}$ **do**
  $x \leftarrow x + e^{-\lambda t}\frac{(\lambda t)^n}{n!} \quad n \leftarrow n+1$
end
return n

We can thus compute the Call prices $C^j(S_0, \sigma)$ based on the stock prices dynamics that include the jump processes. The parameters that change are the moneyness, i.e the ratio

of the strike price K and the initial stock price S_0, and the maturity time T. We focus on $K/S_0 \in \{0.8, 0.9, 1.0, 1.1\}$ and $T \in \{0.02, 0.08, 0.25, 0.5\}$ The results are given in the following table.

Maturity	Strike 0.8	Strike 0.9	Strike 1.0	Strike 1.1
0.02	20.0640	10.0733	1.1802	0.0003
0.08	20.2558	10.3573	2.4489	0.1338
0.25	20.8335	11.5192	4.5488	1.1774
0.5	21.8287	13.2162	6.7183	2.8307

TABLE 1: Call prices matrix by maturity and strike

Our aim is to find the implied volatility of this stocks, which is by definition the volatility σ^{IV} such that $C^{BS}(S_0, K, \tau, \sigma^{IV}) = C^j(S_0, K, \tau, \sigma)$. This can be done numerically using `brentq`[1] module from `scipy.stats`. `brentq` is used to compute roots of functions, so the function passed to the algorithm is $f(\sigma^{IV}) = C^{BS}(S_0, K, \tau, \sigma^{IV}) - C^j(S_0, K, \tau, \sigma)$.

We then get the following four plots of the Black-Scholes implied volatility of the call as a function of strike (moneyness) K/S using (BS):

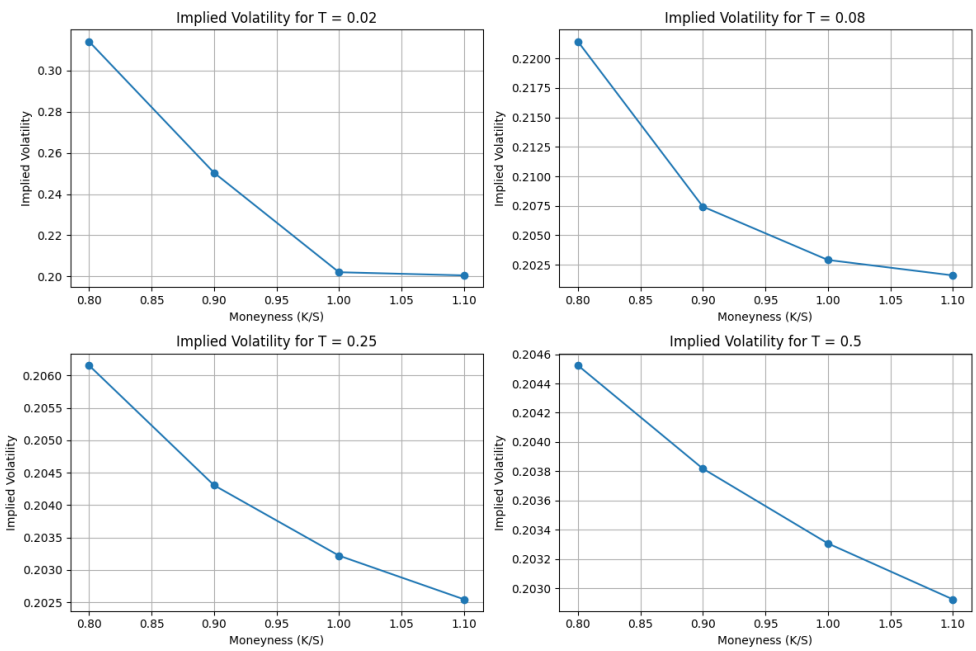

FIG. 1: Implied Volatilities vs Moneyness for different maturities

[1]Brent, R. P., Algorithms for Minimization Without Derivatives. Englewood Cliffs, NJ: Prentice-Hall, 1973. Ch. 3-4.